# Do Financial Inputs Drive League Success? Panel Evidence from Six Premier League Clubs

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ECO375: Applied Econometrics
University of Toronto
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December 4th, 2025

Abstract: The rapid commercialisation of professional football, driven principally by the expansion of broadcasting markets, has intensified interest in understanding how financial capacity shapes competitive outcomes. In the English Premier League (EPL), where revenue disparities have grown markedly, identifying the channels through which financial resources translate into on-field performance is central to debates concerning competitive balance, regulatory design, and the sustainability of club business models. Motivated by the need for rigorous empirical evidence to inform policy discussions, such as revenue sharing, Financial Fair Play regulation, and investment incentives for clubs, this study examines the link between financial strength and league performance.

We analyse six major EPL clubs, Arsenal, Chelsea, Everton, Manchester City, Manchester United, and Tottenham Hotspur, across the 2004–2024 period, using a panel of 105 club-season observations derived from audited financial reports. The empirical design estimates the association between league points and several financial indicators: broadcasting revenue, non-broadcasting revenue, wage expenditure, shareholders' funds, and UEFA qualification. Simple OLS regressions indicate strong positive correlations between financial inputs and performance, with broadcasting and non-broadcast revenues showing particularly large

coefficients. Fixed-effects estimates reveal attenuation of most coefficients when time-invariant club characteristics are accounted for, suggesting that much of the observed correlation reflects structural differences rather than short-run financial changes. Random-effects models nonetheless show that broadcasting revenue and UEFA qualification remain significant predictors of performance.

The results highlight the role of persistent structural heterogeneity, most notably the sustained outperformance of Manchester City and the underperformance of Everton, in shaping long-run outcomes. The study contributes to the sports economics literature by providing a financially disaggregated analysis of performance determinants and by offering evidence relevant to ongoing policy debates. While estimates are descriptive rather than causal, they underscore how financial resources and institutional characteristics jointly structure competitive dynamics in elite football.

## 1. Introduction

Professional football has undergone substantial commercial expansion over the past two decades, driven largely by rapidly escalating broadcasting rights. As the English Premier League (EPL) has become the world's most financially powerful domestic competition, revenues from domestic and international media deals increasingly shape clubs' spending, squad investment, and competitiveness. This raises a central question in sports economics: to what extent does financial strength translate into superior on-field performance?

Prior research documents strong associations between total revenues, wages, and sporting outcomes, but far less is known about the relative importance of specific revenue components, particularly broadcast income, which has grown disproportionately relative to commercial and matchday streams. The role of longer-term financial stability, measured through shareholders' funds, remains theoretically ambiguous.

This study evaluates the relationship between financial inputs and domestic league performance for Arsenal, Chelsea, Everton, Manchester City, Manchester United, and Tottenham Hotspur, over 2004–2024. A 105-observation panel combining audited financial accounts and season-level league results is used to assess how broadcast revenue, non-broadcast revenue, wages, shareholder funds, and UEFA qualification relate to league points.

The empirical strategy proceeds by estimating simple OLS regressions (Tables 1–5) and then multivariate fixed-effects and random-effects models that account for club-specific heterogeneity. Predictive margins (Graphs 1–2) illustrate how expected points vary across revenue levels and clubs. Simple OLS estimates show strong correlations between financial strength and league performance. Coefficients weaken in FE and RE models, indicating that much of the OLS relationship reflects persistent structural differences rather than short-term financial variation. Nonetheless, predictive margins demonstrate economically meaningful performance gaps aligned with financial capacity. These findings have implications for debates on competitive balance, revenue sharing, and financial regulation, and contribute to empirical research on the financial determinants of on-field success.

Based on the existing literature and institutional context, the analysis proceeds with the following hypotheses:

H1: Higher broadcasting revenue is associated with higher league points; H2: Non-broadcast revenue positively predicts league performance, though with a smaller effect; H3: Wage expenditure is positively associated with league points; H4: Shareholders' funds are positively related to performance, although the theoretical sign is ambiguous; H5: UEFA qualification status is strongly positively associated with league points; H6: These relationships attenuate in multivariate regressions once club-specific characteristics are controlled for.

## 2. The Context and Data

The dataset consists of 105 club-season observations for Arsenal, Chelsea, Everton, Manchester City, Manchester United, and Tottenham over 2004–2024. The dependent variable is league points; predictors include log(broadcasting revenue), log(non-broadcasting revenue), total wages, log(shareholders' funds), and a UEFA qualification indicator (1 = Champions League; 0.5 = Europa League; 0 = neither). Each observation is linked to a club identifier, allowing econometric models to account for persistent differences in club size, brand strength, and ownership structure.

Financial variables were derived from audited annual reports; reconstructed disaggregated figures were used only where unavailable (notably early Chelsea seasons). Substantial variation exists in both revenues and performance: Everton and pre-investment Manchester City exhibit lower point totals, while Manchester United, Chelsea, and post-2010 Manchester City consistently rank higher. Summary statistics (Table A) illustrate wide dispersion in commercial income and shareholders' funds and strong temporal growth in broadcast revenue, providing sufficient cross-sectional and temporal variation for regression analysis.

The empirical strategy proceeds in three steps: (i) OLS regressions assess pairwise associations between financial variables and league points; (ii) fixed-effects regressions test whether within-club financial fluctuations predict performance; and (iii) random-effects regressions combine within- and between-club variation while retaining time-invariant structure. Standard errors are clustered by club, and predictive margins illustrate substantive effects. This dataset therefore, offers an appropriate structure for evaluating the financial mechanisms shaping competitive outcomes.

## 3. Regression Analysis

$$Points_{it}=\beta_1 log(Broadcast_{it})+\beta_2 log(NonBroadcast_{it})+\beta_3 Wages_{it}+\beta_4 log(Funds_{it})+\beta_5 UEFA_{it}+\alpha_i+\varepsilon_{it}$$

where i indexes clubs and t seasons, $\alpha_i$ captures time-invariant club characteristics, and errors $\varepsilon_{it}$ are clustered by club.

### 3.1 Simple OLS Regressions (Tables 1–5)

Broadcasting Revenue (Table 1)

The coefficient on log(broadcast revenue) is 6.555 ($p = 0.002$), indicating a strong and statistically significant relationship. A one-unit log increase predicts approximately 6.6 additional points, with a 95% CI of [2.5, 10.6].

Shareholders' Funds (Table 2)

The coefficient of 7.152 ($p = 0$) suggests financial stability strongly correlates with performance. Clubs with higher equity consistently perform better.

Wages (Table 3)

The wage coefficient is positive (0.044, $p = 0.01$), supporting the well-established finding that squad investment predicts performance.

Non-Broadcast Revenue (Table 4)

Non-broadcast revenue shows a strong and significant positive coefficient (8.553, $p = 0$), consistent with long-run financial capacity contributing to sporting success.

UEFA Qualification (Table 5)

UEFA qualification has the largest coefficient (26.396, $p = 0$), reflecting the strong alignment between elite club status and performance.

Taken collectively, the OLS estimates reveal a robust pattern in which financially stronger clubs outperform their peers and meet hypotheses 1-5, although these relationships must be interpreted

cautiously given their purely associative nature. These patterns are consistent with prior findings in the football economics literature, particularly the strong wage–performance and revenue–performance relationships documented by Szymanski (1998, 1999), Forrest and Simmons (2002), and Dawson et al. (2000).

### 3.2 Fixed-Effects Regression (Table 6)

The FE model indicates that after removing all time-invariant club characteristics, broadcasting revenue remains economically meaningful ($\beta \approx 7.97$) but loses statistical significance ($p \approx 0.062$). Non-broadcast revenue, wages, and shareholder funds show weakened coefficients and are not statistically significant. This implies that much of the variation in performance explained by financial variables in OLS reflects between-club differences, not within-club changes. UEFA qualification is the only variable that remains strongly significant within the FE model. A Hausman test (not shown) was used to compare the fixed-effects and random-effects estimators; the results support presenting both specifications for comparative interpretation.

### 3.3 Random-Effects Regression (Table 7)

The random-effects model, which incorporates both within- and between-club variation, yields more efficient estimates given the small number of clubs. Log broadcasting revenue remains positive and statistically significant, indicating that media-rights income is a consistent predictor of league performance even after accounting for club-level heterogeneity. UEFA qualification also remains strongly significant, reflecting its role as a proxy for underlying squad strength. The club indicators reveal marked structural differences: Manchester City outperforms the baseline by about fifteen points, while Everton underperforms by roughly nine. These patterns highlight the influence of long-run structural advantages such as ownership resources, commercial scale, and institutional stability on persistent performance gaps across clubs. Taken together, the FE and RE results are consistent with H6: once persistent club heterogeneity is accounted for, the apparent strength of financial predictors declines, indicating that much of the OLS relationship reflects structural differences rather than within-club financial variation.

### 3.4 Predictive Margins (Graphs 1–2)

Graph 1: Margins for log_broadcast revenue

Predicted points rise from about 110 at log 10 to 150 at log 15, confirming that higher broadcast income is associated with substantially better performance.

Graph 2: Margins by club (RE model)

The random-effects margins reveal substantial and persistent performance differences across clubs. Predicted points are highest for Manchester City (≈85), followed by Chelsea (≈74), Manchester United (≈71), Arsenal (≈69), and Tottenham (≈67). Everton records the lowest expected total (≈59). These patterns align with long-run financial strength and indicate that structural advantages, including ownership resources, commercial scale, and sustained investment capacity, continue to shape league outcomes even after accounting for financial covariates.

## 4. Limitations

Several limitations constrain the interpretation of the findings. First, the analysis cannot establish causality. Many unobserved factors, managerial quality, tactical systems, injury shocks, age structure, and recruitment efficiency, likely influence both financial variables and on-field success, generating omitted-variable bias. In this setting, the conditional mean independence assumption does not hold, so the results should be understood as descriptive correlations rather than causal estimates. Second, financial variables may be endogenous. Broadcasting and commercial revenues can increase following strong performance, creating potential reverse causality. Third, reconstructed revenue splits for earlier years, especially pre-2010 Chelsea values, introduce measurement error that may attenuate estimated effects. Fourth, the small number of clubs (N = 6) reduces statistical power, particularly in fixed-effects regressions, and may obscure genuine within-club relationships. Finally, external validity is limited. The sample consists exclusively of financially large EPL clubs operating under similar structural and regulatory conditions; results may not generalize to smaller teams, lower divisions, or leagues with different revenue-sharing arrangements. These concerns jointly limit both internal validity (due to simultaneity and omitted factors) and external validity (due to the elite nature of the clubs studied).

## 5. Conclusion

Across multiple empirical specifications, financial health shows a strong and economically meaningful relationship with league performance. Broadcasting revenue, non-broadcast revenue, wages, shareholder funds, and UEFA qualification all exhibit positive and significant correlations with league points in simple regressions. While multivariate FE and RE models weaken the statistical significance of some financial predictors, the economic magnitudes remain substantial, and predictive margins show clear upward trends in expected performance with higher financial inputs.

Overall, results support the conclusion that financial well-being is a central determinant of competitive success in the English Premier League. Clubs with strong revenue bases and equity positions sustain higher performance levels, reinforcing the broader literature in sports economics. Future work incorporating larger samples or instrumental-variable strategies would strengthen causal interpretation, but this study provides compelling descriptive evidence that financial resources and on-field success remain deeply interconnected.

| Table A: Summary Statistics | | | | |
|---|---|---|---|---|
| **Variable** | **Observations** | **Mean** | **SE** | **[Min, Max]** |
| Points | 105 | 69.057 | 13.93 | [36, 100] |
| log_broadcast | 105 | 4.665 | 0.6396 | [3.033, 5.702] |
| log_non_broadcast | 105 | 4.96 | 0.8223 | [3.174, 6.087] |
| log_funds | 105 | 5.669 | 0.751 | [3.714, 6.872] |
| wages | 105 | 182.151 | 100.3014 | [30.84, 422.895] |
| uefa_qual | 105 | 0.648 | 0.4156 | [0, 1] |

| Table 1: Simple regression of League Points on Broadcasting Revenue (log) | | | | | |
|---|---|---|---|---|---|
| R-squared | 0.0906 | | | | |
| **Points** | **Coefficient** | **Robust SE** | **t** | **P > \| t \|** | **95% CI** |
| log_broadcast | 6.555 | 2.046 | 3.20 | 0.002 | [2.5, 10.613] |
| y-intercept | 38.477 | 9.635 | 3.99 | 0 | [19.37, 57.59] |

| Table 2: Simple regression of League Points on Shareholders' Funds (log) | | | | | |
|---|---|---|---|---|---|
| R-squared | 0.1486 | | | | |
| **Points** | **Coefficient** | **Robust SE** | **t** | **P > \| t \|** | **95% CI** |
| log_funds | 7.152 | 1.687 | 4.24 | 0 | [3.8, 10.497] |
| y-intercept | 28.508 | 9.644 | 2.96 | 0.004 | [9.38, 47.64] |

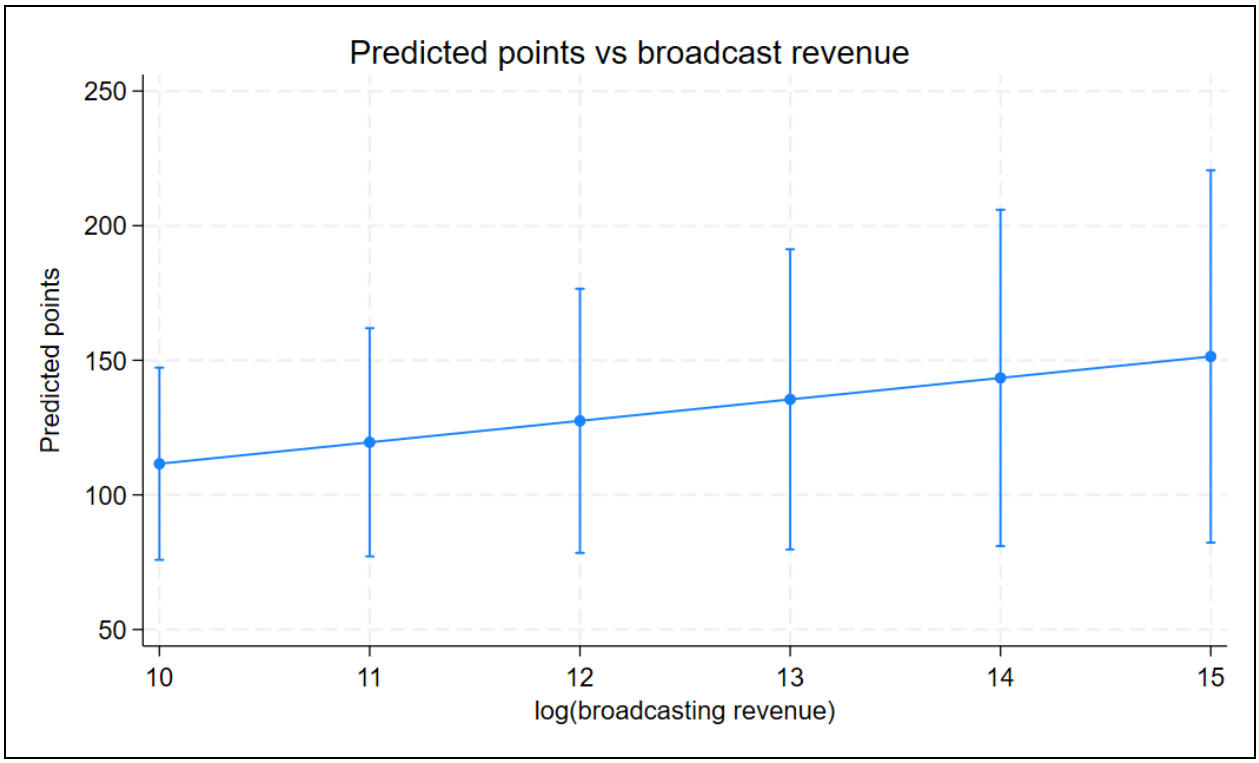
| Table 3: Simple regression of League Points on Wages | | | | | |
|---|---|---|---|---|---|
| R-squared | 0.1017 | | | | |
| **Points** | **Coefficient** | **Robust SE** | **t** | **P > \| t \|** | **95% CI** |
| wages | 0.044 | 0.1297 | 3.41 | 0.01 | [0.0186, 0.07] |
| y-intercept | 60.99 | 2.69 | 22.64 | 0 | [55.65, 66.33] |

| Table 4: Simple regression of League Points on Non-Broadcasting Revenue (log) | | | | | |
|---|---|---|---|---|---|
| R-squared | 0.255 | | | | |
| **Points** | **Coefficient** | **Robust SE** | **t** | **P > \| t \|** | **95% CI** |
| log_non_broadcasting | 8.553 | 1.44 | 5.94 | 0 | [5.696, 11.41] |
| y-intercept | 26.632 | 7.2427 | 3.68 | 0 | [12.27, 40.996] |

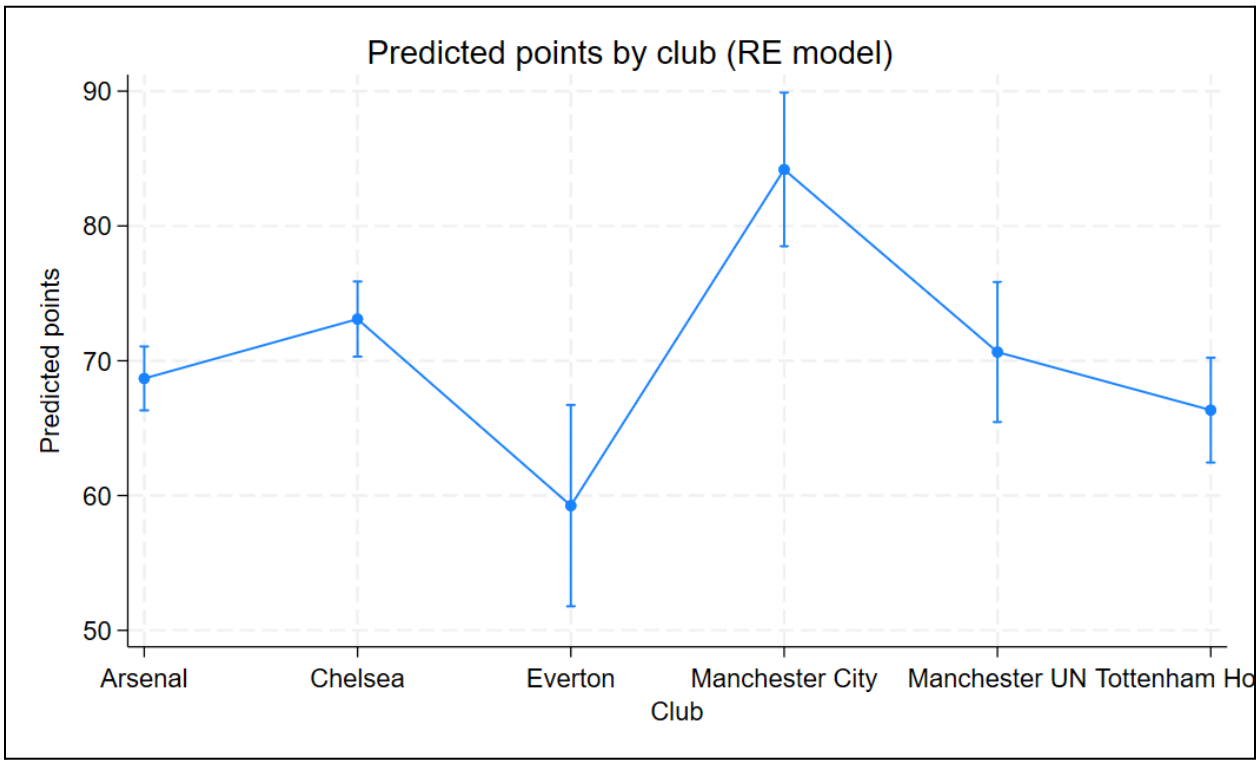
| Table 5: Simple regression of League Points on UEFA Qualification | | | | | |
|---|---|---|---|---|---|
| R-squared | 0.6202 | | | | |
| **Points** | **Coefficient** | **Robust SE** | **t** | **P > \| t \|** | **95% CI** |
| uefa_qual | 26.396 | 2.035 | 12.97 | 0 | [22.36, 30.43] |
| y-intercept | 51.96 | 1.564 | 33.23 | 0 | [48.86, 55.06] |

| Table 6: Fixed-Effects Regression of League Points on Revenue Components and Financial Controls | | | | | |
|---|---|---|---|---|---|
| sigma_u | 8.259 | | | | |
| sigma_e | 7.865 | | | | |
| rho | 0.524 | | | | |
| R-squared | 0.4867 | | | | |
| **Points** | **Coefficient** | **Robust SE** | **t** | **P > \| t \|** | **95% CI** |
| log_broadcast | 7.968 | 3.33 | 2.4 | 0.062 | [-0.581, 16.518] |
| log_non_broadcast | 1.526 | 2.868 | 0.53 | 0.617 | [-5.845, 8.898] |
| wages | -0.503 | 0.0399 | -1.26 | 0.263 | [-.153, 0.052] |
| uefa_qual | 18.92 | 1.749 | 10.82 | 0 | [14.424, 23.417] |
| log_funds | -5.193 | 2.525 | -2.06 | 0.095 | [-11.683, 1.297] |
| y-intercept | 50.6687 | 17.638 | 2.87 | 0.035 | [ 5.33, 96.008] |

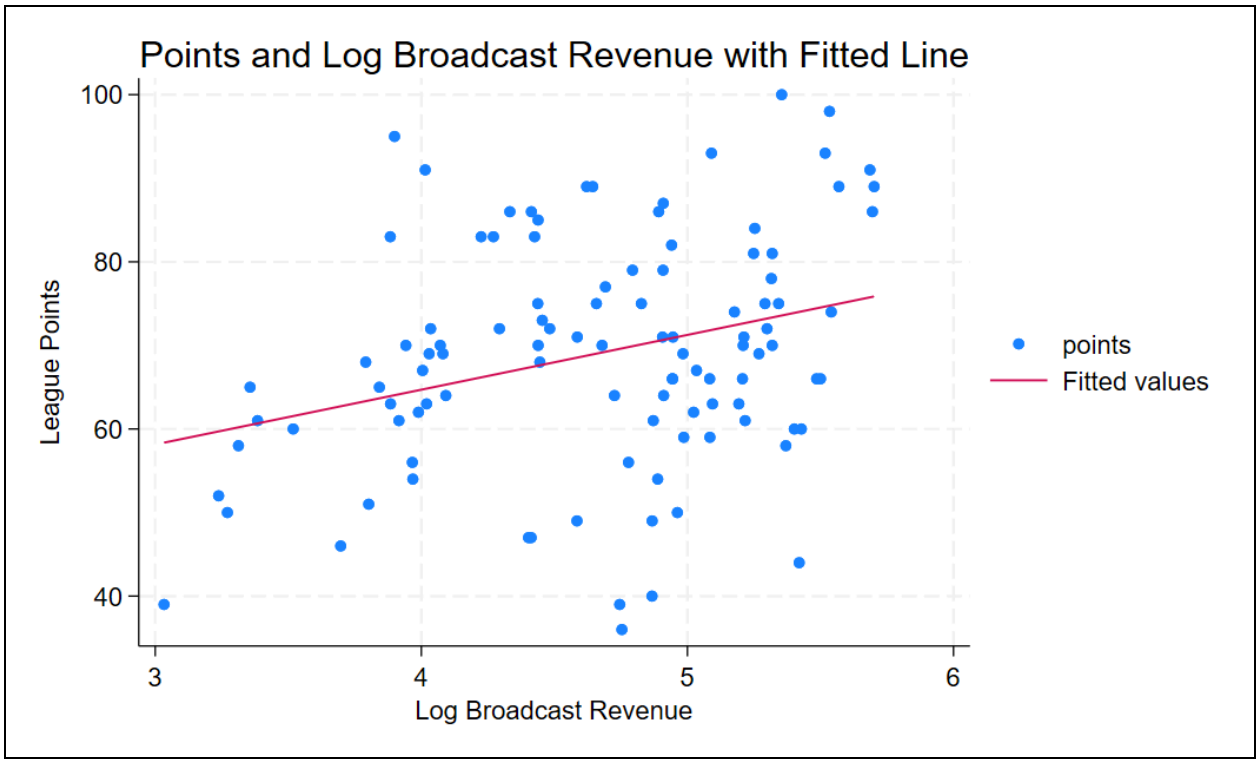
| Table 7: Random-Effects Regression of League Points on Revenue Components, Financial Controls, and Club Indicators | | | | | |
|---|---|---|---|---|---|
| sigma_u | 0 | | | | |
| sigma_e | 7.865 | | | | |
| rho | 0 | | | | |
| R-squared | 0.7119 | | | | |
| **Points** | **Coefficient** | **Robust SE** | **z** | **P > \| z \|** | **95% CI** |
| log_broadcast | 7.968 | 3.413 | 2.33 | 0.2 | [1.278, 14.658] |
| log_non_broadcast | 1.526 | 2.943 | 0.52 | 0.604 | [-4.242, 7.294] |
| wages | -0.0503 | 0.0409 | -1.23 | 0.219 | [-0.131, 0.0299] |
| uefa_qual | 18.92 | 1.795 | 10.54 | 0.00 | [15.402, 22.439] |
| log_funds | -5.193 | 2.59 | -2.00 | 0.045 | [-10.272, -0.115] |
| **club_id** | | | | | |
| Chelsea | 4.41 | 2.23 | 1.98 | 0.048 | [0.035, 8.78] |
| Everton | -9.43 | 4.57 | -2.06 | 0.039 | [-18.392, -0.474] |
| Manchester City | 15.499 | 3.2 | 4.84 | 0 | [9.227, 21.772] |
| Manchester United | 1.958 | 2.899 | 0.68 | 0.499 | [-3.724, 7.639] |
| Tottenham Hotspurs | -2.347 | 1.252 | -1.88 | 0.061 | [-4.8, 0.106] |
| y-intercept | 50.2969 | 19.20788 | 2.62 | 0.009 | [12.65, 87.944] |



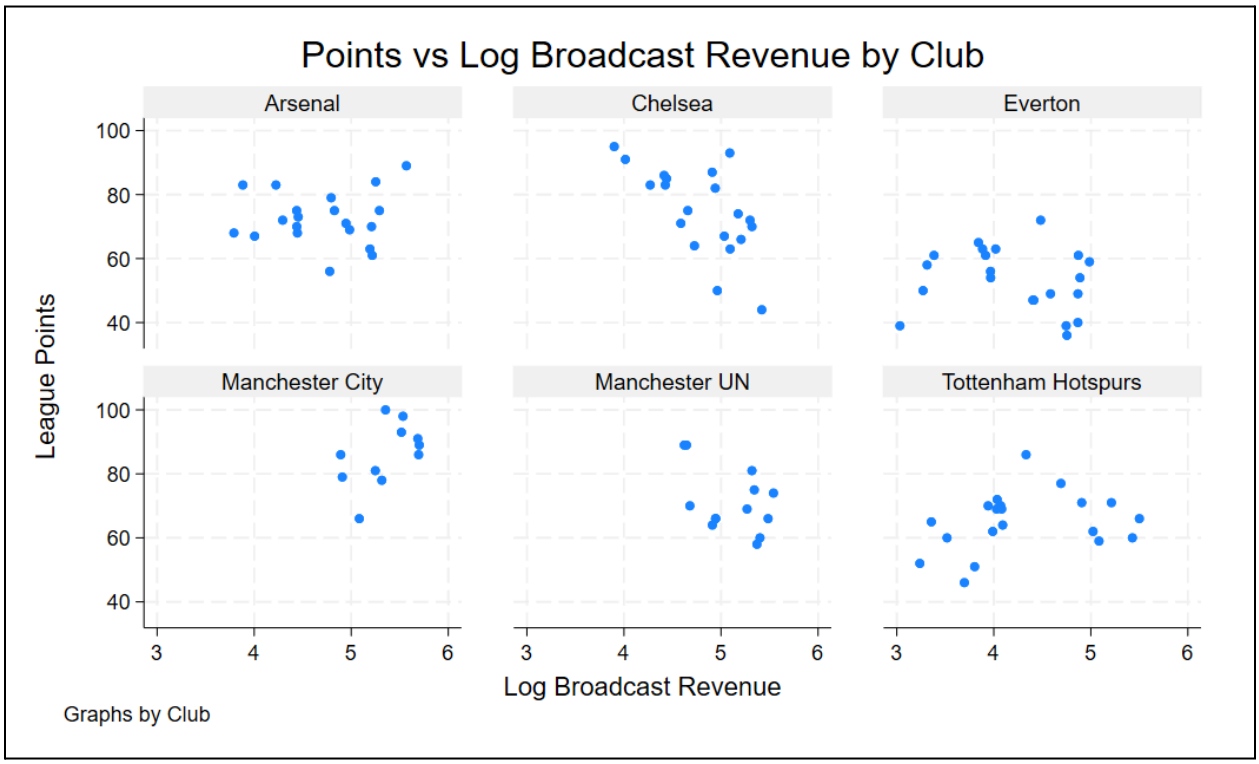

Graph 1: Predicted League Points Across Levels of Log Broadcast Revenue




Graph 2: Predicted points by club (RE model)




Graph 3: Scatterplot of points and broadcast revenue (log)




Graph 4: Scatterplot of points and broadcast revenue (log), by club

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